EC2 — Important Interview Questions & Answers

1.What is Amazon EC2?
Amazon EC2 is a cloud computing service in AWS used to launch and run virtual servers called instances.
It allows us to choose the required compute capacity, operating system, storage, and networking configuration.
Simple ga gurthupettuko:
EC2 ante AWS lo virtual servers ni create and run cheyyadaniki use chese service.
Eng: EC2 is a service used to launch and run virtual servers in AWS.

2.What is an EC2 Instance?
An EC2 instance is a virtual server running in the AWS cloud.
It provides compute capacity to run applications, websites, and other workloads.
Simple ga gurthupettuko:
EC2 Instance ante AWS lo run ayye virtual server.
Eng: An EC2 instance is a virtual server running in AWS.

3.What are EC2 Instance Types?
EC2 instance types define the combination of CPU, memory, storage, and network performance available for an instance.
Different instance families are designed for general purpose, compute, memory, storage, or accelerated workloads.
Simple ga gurthupettuko:
Instance Type ante EC2 ki entha CPU, memory and performance kavalo choose chese option.
Eng: Instance Types define the compute, memory, and performance capacity of an EC2 instance.

4.What is an AMI?
AMI stands for Amazon Machine Image.
It contains the information required to launch an EC2 instance, such as the operating system, applications, and configuration.
Simple ga gurthupettuko:
AMI ante EC2 instance launch cheyyadaniki use chese machine image.
Eng: An AMI contains the information required to launch an EC2 instance.

5.What is a Key Pair in EC2?
A Key Pair is used to securely connect to an EC2 instance.
It consists of a public key stored by AWS and a private key that the user keeps securely.
Simple ga gurthupettuko:
Key Pair ante EC2 ki secure ga connect avvadaniki use chese key.
Eng: A Key Pair is used to securely connect to an EC2 instance.

6.What is EC2 User Data?
User Data is a script or set of commands that can run when an EC2 instance is launched.
It is commonly used for initial configuration and installing required software.
Simple ga gurthupettuko:
User Data ante EC2 launch ayye time lo initial setup kosam run chese script.
Eng: User Data is used to run initialization scripts when an EC2 instance launches.

7.What is EBS in EC2?
Amazon EBS provides persistent block storage for EC2 instances.
It can store operating systems, applications, and data, and the data can persist independently of the instance lifecycle depending on configuration.
Simple ga gurthupettuko:
EBS ante EC2 ki persistent storage provide chese AWS service.
Eng: EBS provides persistent block storage for EC2 instances.

8.What is Instance Store?
Instance Store is temporary block storage physically attached to the host computer of an EC2 instance.
Its data is lost when the instance is stopped, hibernated, or terminated, depending on the instance lifecycle.
Simple ga gurthupettuko:
Instance Store ante EC2 ki temporary storage, instance lifecycle change ayite data lose avvachu.
Eng: Instance Store provides temporary storage that is tied to the EC2 instance lifecycle.

9.What is the difference between EBS and Instance Store?
EBS is persistent storage that can continue to exist independently of an EC2 instance depending on its configuration.
Instance Store is temporary storage physically attached to the host and is lost when the instance is stopped or terminated.
Simple ga gurthupettuko:
EBS persistent storage, Instance Store temporary storage.
Eng: EBS provides persistent storage, while Instance Store provides temporary storage.

10.What are Placement Groups in EC2?
Placement Groups are used to control how EC2 instances are placed on the underlying AWS hardware.
The main types are Cluster, Spread, and Partition placement groups.
Simple ga gurthupettuko:

Placement Group ante EC2 instances physical ga ela place avvali ani control cheyyadaniki use chestham.
Eng: Placement Groups control the placement of EC2 instances on AWS hardware.

11.What is a Cluster Placement Group?
A Cluster Placement Group places instances close together in the same Availability Zone.
It is useful for workloads that require high network performance and low network latency between instances.
Simple ga gurthupettuko:
Cluster Placement Group ante instances ni close ga place chesi high network performance kosam use chestham.
Eng: A Cluster Placement Group places instances close together for high network performance.

12.What is a Spread Placement Group?
A Spread Placement Group places instances across distinct underlying hardware.
It helps reduce the risk of simultaneous hardware failure affecting multiple instances.
Simple ga gurthupettuko:
Spread Placement Group ante instances ni different hardware meeda spread cheyyadam.
Eng: A Spread Placement Group spreads instances across separate hardware.

13.What is a Partition Placement Group?
A Partition Placement Group divides instances into logical partitions with separate underlying hardware.
It is useful for large distributed workloads where failure isolation is important.
Simple ga gurthupettuko:
Partition Placement Group ante instances ni separate partitions and hardware lo place cheyyadam.
Eng: A Partition Placement Group separates instances into partitions for failure isolation.

14.What is a Dedicated Host?
A Dedicated Host is a physical server fully dedicated to your AWS account.
It gives you visibility and control over the physical host and can be useful for specific licensing or compliance requirements.
Simple ga gurthupettuko:
Dedicated Host ante complete physical server mana AWS account kosam dedicated ga undadam.
Eng: A Dedicated Host is a physical server dedicated to a single AWS account.

15.What is a Dedicated Instance?
A Dedicated Instance runs on hardware dedicated to a single AWS account.
However, unlike a Dedicated Host, it does not provide the same level of visibility and control over the physical host.
Simple ga gurthupettuko:
Dedicated Instance ante dedicated hardware meeda run ayye EC2 instance.
Eng: A Dedicated Instance runs on hardware dedicated to a single AWS account.

16.What is the difference between Dedicated Host and Dedicated Instance?
A Dedicated Host provides a dedicated physical server with visibility and control over the host.
A Dedicated Instance runs on dedicated hardware but does not provide the same physical host-level control.
Simple ga gurthupettuko:
Dedicated Host lo physical server meeda control ekkuva, Dedicated Instance lo control limited.
Eng: Dedicated Hosts provide more control over dedicated physical hardware than Dedicated Instances.

17.What happens when an EC2 instance is stopped?
When an EC2 instance is stopped, the compute resources are released and the instance can be started again later.
Data on attached EBS volumes is normally preserved, while instance store data is lost.
Simple ga gurthupettuko:
EC2 stop cheste compute release avtundi, EBS data normally untundi, Instance Store data lose avvachu.
Eng: When an EC2 instance is stopped, compute resources are released while EBS data is normally preserved.

18.What happens when an EC2 instance is terminated?
When an EC2 instance is terminated, the instance is permanently deleted.
EBS volumes configured for deletion on termination are also deleted, while volumes configured to persist can remain.
Simple ga gurthupettuko:
Terminate ante EC2 permanently delete avvadam.
Eng: Terminating an EC2 instance permanently deletes the instance.

19.How would you choose an EC2 instance type?
I would first understand the application's CPU, memory, storage, network, and performance requirements.
Then I would select an appropriate EC2 instance family and size based on the workload.
Simple ga gurthupettuko:
Application requirement batti CPU, memory and performance ki suitable Instance Type choose chestham.
Eng: I would choose an EC2 instance type based on the workload requirements.

20.How would you troubleshoot an EC2 instance that is not reachable?
I would check the instance state, security group, network ACL, route table, subnet, Internet Gateway or NAT configuration, and the instance's operating system.
I would also check system and instance status checks and relevant logs.
Simple ga gurthupettuko:

EC2 reachable kakapothe State → Security Group → NACL → Route Table → Subnet → OS check chestham.
Eng: I would check the instance state, networking, security rules, and operating system.

21.What is the most important thing to remember about EC2?
EC2 is mainly used to provide scalable virtual compute capacity in AWS.
The important concepts are Instances, Instance Types, AMIs, Key Pairs, User Data, EBS, Instance Store, Placement Groups, and Dedicated Hosts.
Simple ga gurthupettuko:
EC2 ante AWS lo scalable virtual servers ni run cheyyadaniki use chese compute service.
Eng: EC2 is a scalable compute service used to run virtual servers in AWS.